# Notes App

Create, view, edit, search and delete personal notes in a native Android app with a custom RecyclerView adapter and SQLite persistence.

## Requirement coverage

**Lists and models** - Entry models, a custom ListAdapter, DiffUtil and ViewHolder render the note list. Stable database IDs keep records distinct even when titles match.

**Local CRUD** - SQLiteOpenHelper stores title and full text. A single background executor serializes reads and writes. Viewing opens the complete note; deletion requires confirmation.

**Input and lifecycle** - Title: 1-120 trimmed characters. Note body: required, up to 10,000 characters. Search is literal and case-insensitive. Editor drafts and in-flight save state survive Activity recreation.

## Run and review

Open the source folder in Android Studio, install SDK 35 / Build Tools 35.0.0 and use JDK 17 or 21. Sync Gradle and run the notes module on Android 8.0 / API 26 or later.

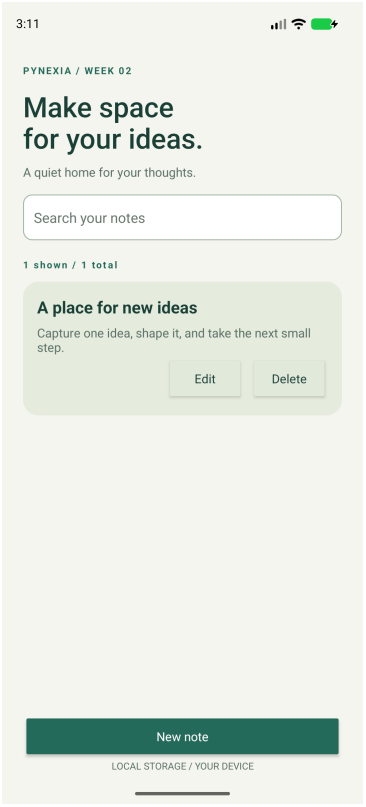
1. Tap New note. Try saving empty fields, then enter a title and body.
2. Save and select the note to read the full content.
3. Edit the note, search for part of its title or body, and reopen the app.
4. Cancel a deletion to keep the note, then confirm deletion to remove it.

## Scope

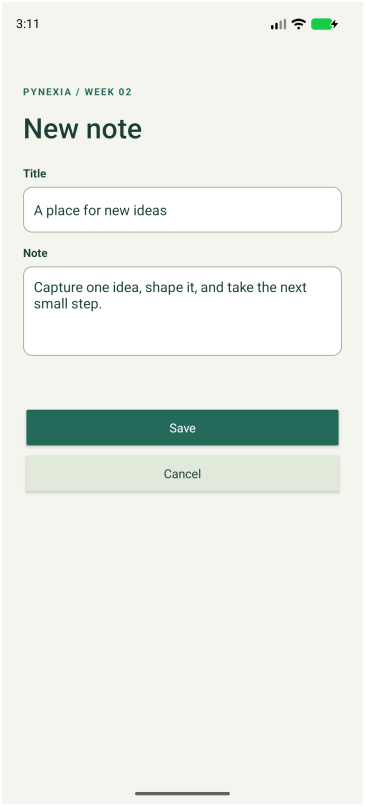
Local storage only. No cloud sync or encryption. Clearing app data/uninstalling removes notes. Duplicate titles are allowed; records have distinct IDs.

# Actual app screens

Captured from the tested native app on the emulator.




Saved note in RecyclerView




Note editor

## Executed verification

4 JVM tests + 3 device tests passed. Lint: 0 errors, 18 warnings. No failures, errors or skipped tests.

Validation boundaries, literal search, SQLite CRUD and close/reopen persistence, UI add/edit/search/delete, cancellation and editor recreation.

Pixel 8 emulator, Android 17 / API 37 (37.1 image), x86_64. No physical-device or API 26 test run. Lint warnings are retained without suppression. APKs are debug-signed evaluation builds.

## Reproduce

.\gradlew.bat :notes:assembleDebug :notes:testDebugUnitTest :notes:lintDebug

.\gradlew.bat :notes:connectedDebugAndroidTest

The second command needs a running emulator or connected device. Initial Gradle sync needs internet. See verification/ for executed test results and lint findings.

# Explanation & submission

يتيح تطبيق الملاحظات إضافة عنوان ونص، وعرض الملاحظة كاملة وتعديلها وحذفها مع تأكيد الحذف. تظهر البيانات في قائمة بمحول مخصص وتحفظ في قاعدة بيانات محلية. يدعم البحث في العنوان والنص، ويحافظ على مسودة التحرير عند تدوير الشاشة.

## Repository URL

https://github.com/ENGEyad/pynexia-week2-notes

## Live Demo / Hosted URL

https://engeyad.github.io/pynexia-week2-notes/

A hosted project showcase with real screenshots and downloads, not an Android emulator.

## Attach File / Archive

Notes-App.zip - source, Gradle wrapper, APK, screenshots, verification and this PDF. Below the 10 MB limit. APK is inside the ZIP because the earlier submission form does not list direct APK uploads.

## Notes / Implementation Remarks (1334 / 2000 characters)

Native Android Notes App implemented with Java and XML. Create, view, edit, search and delete personal notes in a native Android app with a custom RecyclerView adapter and SQLite persistence.

Entry models, a custom ListAdapter, DiffUtil and ViewHolder render the note list. Stable database IDs keep records distinct even when titles match. SQLiteOpenHelper stores title and full text. A single background executor serializes reads and writes. Viewing opens the complete note; deletion requires confirmation. Title: 1-120 trimmed characters. Note body: required, up to 10,000 characters. Search is literal and case-insensitive. Editor drafts and in-flight save state survive Activity recreation.

Run: open the source folder in Android Studio, sync Gradle, and run notes on Android 8.0/API 26 or later (compile/target SDK 35).

Validation: 4 JVM tests and 3 device tests passed on a Pixel 8 emulator, Android 17/API 37. Lint: zero errors; 18 warnings documented without suppression.

Local storage only. No cloud sync or encryption. Clearing app data/uninstalling removes notes. Duplicate titles are allowed; records have distinct IDs.

Hosted URL: project showcase with actual screenshots and downloads, not a browser emulator. The ZIP contains source, Gradle wrapper, debug APK, screenshots, verification and a bilingual PDF report.